

Smart Learning Assistant System Design and Implementation Report

1. Project Introduction

Smart Learning Assistant is an intelligent learning support system designed for university students. It helps students manage class schedules, learning tasks, course resources, deadline reminders, and short-term study plans in one place. For university students, daily learning information is often scattered: class times may be stored in a timetable, assignments and project tasks may be recorded in a to-do app or chat history, and learning materials may be distributed across web links, documents, and notes. When there are many tasks, students may forget deadlines and may also find it difficult to quickly decide what should be completed first.

The goal of this project is to design and implement an integrated learning management system. With this system, users can view course arrangements, manage tasks, save learning resources, check learning progress, and generate simple study plans according to current tasks and learning goals. The system is not only a to-do list. Instead, it combines schedule management, task management, resource management, reminder functions, and study planning into a more complete learning assistant.

The system uses Vue3 as the front-end framework, Vue Router for page navigation, Element Plus for interface components such as buttons, dialogs, forms, tags, and progress bars, Vite as the build tool, and browser localStorage for storing course, task, resource, and plan data. This technical solution has a clear structure, is easy to run, and is suitable for course project demonstration and local presentation.

The system mainly includes five modules: Dashboard, Schedule, Tasks, Resources, and Planner. These modules are not completely isolated. For example, after the user adds or completes a task in Tasks, the number of pending tasks, completion rate, and deadline reminders in Dashboard will change. After the user adds a course in Schedule, Dashboard can display today's classes. After the user saves learning resources, the resource statistics in Resources and Dashboard will also be updated. This connection between modules makes the system more like a complete learning management platform.

2. Target Users and Requirement Analysis

The target users of this system are university students, especially students who take multiple courses and need to handle several assignments, reading tasks, project tasks, and review plans at the same time. Their learning activities usually have the characteristics of many tasks, fragmented time, many sources of learning materials, and frequent deadlines.

In real learning situations, users may face several problems. First, courses, assignments, resources, and deadlines are often scattered in different places, so students need to switch between several tools. Second, when there are many tasks, students may find it difficult to quickly decide what should be completed first today. Third, assignments and project tasks can be missed, especially when the deadline is close but there is no clear reminder. Fourth, course resource links, reference websites, and learning

notes are not easy to manage in a unified way. Fifth, students sometimes have available time but do not know how to turn that time into a concrete study arrangement.

Based on these problems, the system needs to meet the following user requirements: users need to quickly view today's classes and learning tasks; manage task priority, due date, and completion status; save course-related learning resources; receive reminders for upcoming deadlines; quickly create short-term study plans; and manage courses, tasks, resources, and plans in one system to reduce the burden caused by scattered information.

Therefore, the design focus of Smart Learning Assistant is to help students centralize learning information and use clear interface feedback to help users understand their current learning status and next action.

3. System Function Design

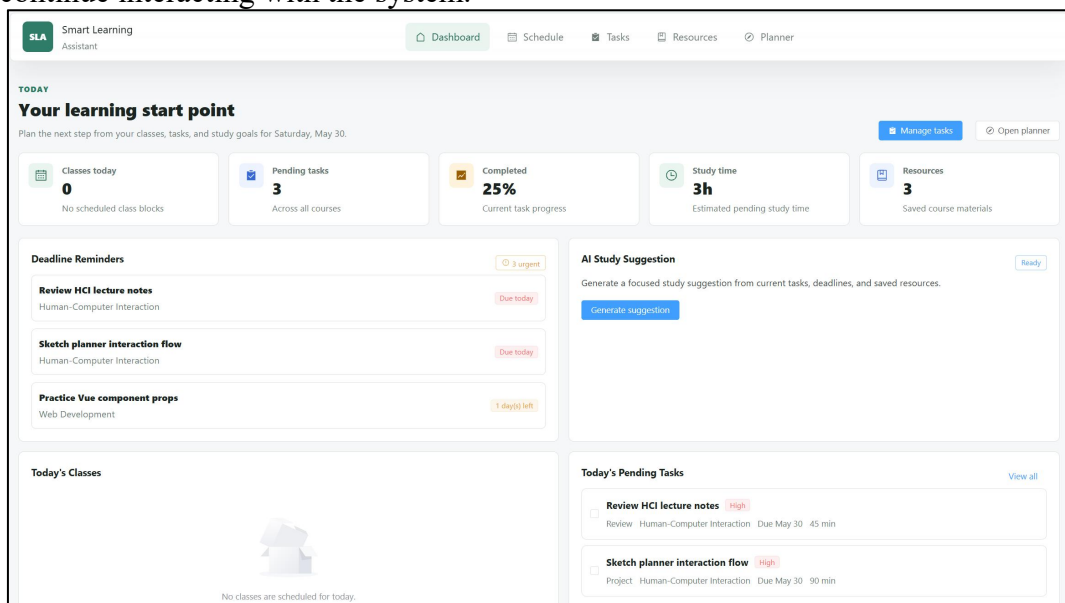
3.1 Dashboard Module

Dashboard is the home page of the system and the first page users see after entering the system. Its main function is to provide an overview of learning status and help users quickly understand current courses, tasks, and learning progress.

Dashboard displays today's classes, the number of pending tasks, task completion rate, estimated study time, upcoming deadlines, course workload, and study suggestions. Through this information, users can quickly judge whether they have classes today, whether there are urgent tasks, how task completion is progressing, and which course has a heavier workload.

Dashboard also provides an AI/local study suggestion function. This function generates the next study suggestion based on current tasks, courses, resources, and deadlines. If the external AI service is unavailable, the system can still generate suggestions through local rules, ensuring stability during demonstration and use.

In addition, some content blocks in Dashboard can be clicked directly. For example, users can click today's class to jump to the Schedule page and further view or edit the course arrangement. This design reduces the time needed to find function entrances and makes Dashboard not only a static display page, but also a central page that guides users to continue interacting with the system.



3.2 Schedule Module

The Schedule module is used to manage weekly course arrangements. The system displays the timetable by week, allowing users to view the distribution of courses across different weekdays. By default, one day contains eleven class periods, including eight daytime periods and three evening periods, which can cover common university course arrangements.

The left side of the timetable contains a period sidebar that shows the period number and its start and end time. Users can click the sidebar to modify the start and end time of a specific period. In this way, the system separates “school timetable settings” from “specific course arrangement”, so users do not need to repeatedly enter exact times whenever they add a course.

When adding a course, users can choose the weekday, start period, and end period. For example, users can directly choose “Monday, Period 1 to Period 2” without manually selecting exact start and end times. After a course is added, it is displayed as a card in the timetable, including course name, location, period, and time information.

The timetable also supports direct interaction. Users can click an empty timetable cell to add a course for the corresponding weekday and period. They can also click an existing course card to edit course information directly. This interaction makes adding and modifying courses more direct and reduces unnecessary operation steps.

WEEKLY VIEW
Class schedule
A simple weekly structure for seeing where study tasks can fit around classes.

Buttons: + Add course, 5 classes

Periods	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Period 1 08:00 - 08:45					Period 1-2 / 08:00 - 09:40 Academic English Room A112		
Period 2 08:55 - 09:40	Period 2-3 / 08:55 - 10:45 Human-Computer Interaction Design Lab 204						
Period 3 10:00 - 10:45			Period 3-4 / 10:00 - 11:40 Web Development Computer Lab 2				
Period 4 10:55 - 11:40							
Period 5 14:00 - 14:45		Period 5-6 / 14:00 - 15:40 Data Visualization Room B305					
Period 6 14:55 - 15:40							
Period 7 16:00 - 16:45							
Period 8 16:55 - 17:40							Period 7-8 / 16:00 - 17:40 Project Review Session Online

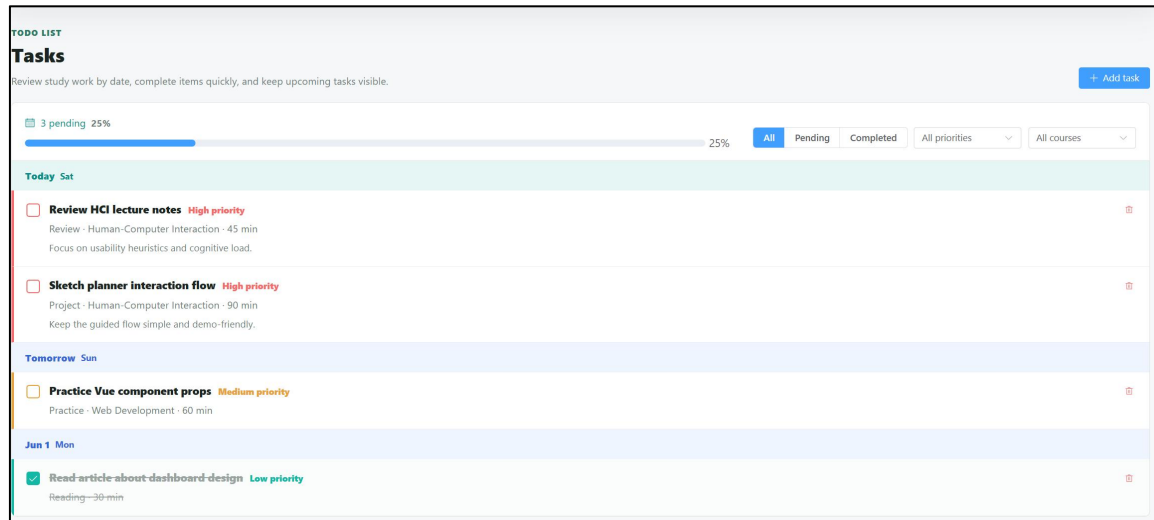
3.3 Tasks Module

The Tasks module is used to manage learning tasks. Users can add, edit, and delete tasks, and can also mark tasks as completed. Each task contains information such as title, related course, task category, priority, due date, estimated study time, and notes.

The task list supports filtering by completion status, priority, and course. When there are many tasks, users can quickly find a specific type of task, such as only unfinished tasks, only high-priority tasks, or only tasks related to a specific course.

Changes in task status affect the data in Dashboard. For example, after the user completes a task, the completion rate in Dashboard increases and the number of pending tasks decreases. If the task has a due date, it also affects upcoming deadline reminders. Through this data connection, users can directly see changes in their learning progress.

The purpose of the Tasks module is to help students clarify current learning tasks and improve task management efficiency through priority, due date, and completion status.



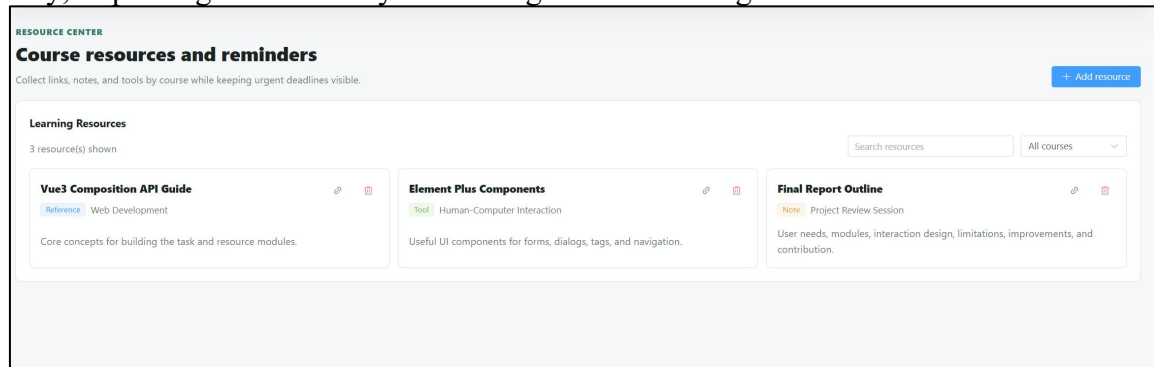
3.4 Resources Module

The Resources module is used to manage learning materials. Users can add course-related resources, such as teaching websites, reference links, online documents, tool links, or learning notes. Each resource can include a title, related course, resource type, link, and note.

Resources can be classified by course and can also be searched and filtered. Users can quickly find resources related to a specific course, avoiding the problem of materials being scattered in browser bookmarks, chat history, or local files.

Resource cards or related buttons can directly open resource links, and users can click the edit button to modify resource information. The page can also show upcoming deadline reminders, so users can notice urgent tasks while checking learning materials.

The Resources module helps students save and find learning materials in a unified way, improving the efficiency of learning resource management.



3.5 Planner Module

The Planner module is used to help users generate short-term study plans. The system guides users through three questions: study subject, available time, and learning

goal. After the user completes the input, the system generates a study plan based on the information and turns a vague learning goal into more concrete steps.

In terms of plan generation, the system supports AI-assisted study plan generation. When the AI service is available, the system sends the user's study subject, available time, and learning goal to a local AI proxy service, and the AI returns a more natural and targeted study plan. If the AI service is unavailable, the system can also generate a basic study plan using local rules, ensuring that the module can still work normally during demonstration and use.

Planner supports browser voice input. In browsers that support speech recognition, users can complete part of the input by speaking. This provides another interaction method and makes the planning process more flexible.

The system can also save recently generated study plans, allowing users to review previous plan content. The purpose of Planner is to help students quickly form an executable study arrangement when their time is limited or their goal is unclear.

4. Interaction Design and User Friendliness

Smart Learning Assistant focuses on clarity, operability, and immediate feedback in interaction design. The system uses a unified navigation structure, so users can quickly switch between Dashboard, Schedule, Tasks, Resources, and Planner without searching for function entrances again.

The interface uses Element Plus components such as buttons, dialogs, tags, progress bars, select controls, and forms, keeping the overall style consistent. Users see similar interaction patterns on different pages, so the learning cost is low. For example, adding courses, tasks, and resources is completed through dialogs or forms, so users do not need to leave the current page to complete major operations.

Dashboard uses cards and progress information to display learning status instead of only listing text. Task completion rate, pending task number, today's classes, and upcoming deadlines are all displayed clearly, making it easy for users to scan. The system centrally displays upcoming deadline tasks in Dashboard, allowing users to quickly identify learning tasks that need priority.

Many content blocks in the system can be clicked to jump or edit directly. For example, users can click today's class in Dashboard to jump to Schedule; click an existing course card to edit course information; click an empty timetable cell to add a course for the corresponding weekday and period; click task content to edit a task; and

click a resource card or button to open a resource link. These designs reduce the distance between “seeing information” and “taking action”.

The Schedule module is an important part of interaction optimization. Users can click the left period sidebar to set the start and end time of each period. When adding a course, users only need to choose “from which period to which period”. Compared with manually selecting exact times every time, this method is closer to school timetable habits and reduces repeated input and possible errors.

The Tasks module provides immediate feedback. After the user completes a task, the system immediately updates the task status, and the completion rate and pending task number in Dashboard also change. Users can directly see the result of their operation, which strengthens feedback.

Planner uses guided questions, allowing users to fill in the study subject, available time, and learning goal step by step. This reduces the pressure of creating a study plan. Voice input provides an additional input method, allowing users to complete plan input by speaking.

In addition, search, filtering, empty-state hints, and color tags improve interface understandability. When there is no data or no filter result, the system shows an empty-state hint, preventing users from thinking that the system has an error. Different colors in tags and course cards also help users distinguish information types more quickly.

5. Project Novelty

The novelty of this project is mainly reflected in system integration, interaction paths, AI assistance, and learning support.

First, the system integrates schedule, tasks, resources, reminders, and study plans into one platform instead of providing only a single function. Users can complete multiple learning management workflows in one system, reducing information fragmentation.

Second, Dashboard not only displays data, but also gives next-step study suggestions based on tasks, courses, and resources. It plays the role of both learning status overview and action guidance.

Third, the timetable uses a period-based design. Users can set the time of each class period first, and then add courses by period. This design better matches university course arrangements and improves course input efficiency.

Fourth, multiple content blocks in the system support direct click-to-jump, add, or edit interactions. For example, course cards, empty timetable cells, today’s classes, and resource content can all trigger follow-up operations, reducing the user’s operation path. Fifth, Planner supports AI-assisted study plan generation. Users only need to enter the study subject, available time, and learning goal, and the system can generate concrete study steps. Even if the AI service is unavailable, the system can still generate a plan using local rules, ensuring functional stability.

Sixth, Planner supports guided plan generation and voice input, allowing users to create study plans in a lighter way.

Seventh, AI suggestion has a local fallback mechanism. Even without an external AI service, the system can still generate local study suggestions, ensuring stability during project demonstration.

6. Program Advantages

The first advantage of the system is functional completeness. It covers course management, task management, resource management, deadline reminders, and study plan generation, forming a relatively complete learning management workflow.

The second advantage is the connection between modules. Tasks, courses, and resources are not isolated; they affect Dashboard statistics and reminder information. This connection improves the integrity of the system.

The third advantage is simple operation. The system uses unified navigation and consistent component styles, so users can quickly understand how to use each module. Operations such as adding, editing, deleting, and filtering are direct.

The fourth advantage is clear interaction feedback. Progress bars, tags, cards, reminders, and dialogs all provide clear feedback. After users complete a task or modify a course, they can immediately see related information changes.

The fifth advantage is that many content blocks can be clicked directly to jump, add, or edit, shortening the operation path. Users do not need to repeatedly search for buttons or switch pages, which improves efficiency.

The sixth advantage is easy deployment. The project can run locally through Vite, and data is stored in localStorage. It does not depend on complex backend configuration and is suitable for course project presentation.

The seventh advantage is good stability. When AI functions are unavailable, the system can still use local suggestion logic, so core functions are not interrupted.

7. Program Limitations

The system still has some limitations.

First, the system does not have a real backend. Data is stored only in the current browser. If users change browsers or devices, the data cannot be synchronized automatically.

Second, the system does not have user account functionality, so it cannot support multi-user login or save independent data for different users.

Third, the AI or recommendation logic is still relatively simple. Current suggestions are mainly based on tasks, courses, and deadlines, and do not perform personalized analysis based on long-term learning behavior.

Fourth, the Resources module can currently only save links and notes. It cannot upload files or directly preview PDFs, courseware, images, or other materials.

Fifth, the system does not have a real notification system. Although upcoming tasks can be displayed inside the application, the system cannot send email, desktop, or system notifications.

Finally, the current system is mainly suitable for course project demonstration and is not yet a complete commercial product. If it is to be used for long-term real-world use, data security, account systems, synchronization mechanisms, and intelligent recommendation capabilities need further improvement.

8. Future Improvements

The system can be improved in the following directions in the future.

First, user login and a database can be added so that different users can have independent accounts and personal data.

Second, cloud synchronization can be supported, allowing users to access their courses, tasks, and resources on different devices.

Third, course timetable import and export can be supported, such as importing courses from standard calendar files or exporting courses and tasks to external calendar software.

Fourth, file upload and material preview can be supported, allowing users to directly manage courseware, PDFs, images, and note files.

Fifth, real notification functions can be added to remind users by email, desktop notifications, or other methods before task deadlines.

Sixth, smarter study plans can be generated based on user learning history, such as analyzing tasks that users often delay, common study times, and course difficulty.

Seventh, richer data visualization can be added, such as weekly study time statistics and task completion statistics by course.

Eighth, group collaboration tasks or shared course resources can be supported, making the system suitable for course project groups or study groups.

9. Team Division and Development Process

9.1 Development Process

At the beginning of the project, the two members discussed the project topic together, identified university students as the target users, and defined the main system modules, including Dashboard, Schedule, Tasks, Resources, and Planner. Then the team designed system functions and page structures based on user requirements.

In the middle stage, the team completed the front-end page structure, routing configuration, local data storage, and core function modules. After the modules were gradually implemented, the team continued to adjust the connections between data so that tasks, courses, resources, and Dashboard could influence one another.

In the final stage, the team optimized interaction details according to the user experience, such as period selection in the timetable, click-to-edit content blocks, task filtering, and Dashboard data feedback. At the same time, the team completed the README, report, and presentation preparation.

9.2 Member Contribution

Member	Main Work	Project Contribution
2452654 郭炫君	Responsible for the overall App layout, routing configuration, Schedule module, Tasks module, and part of the interaction optimization.	50%
2354127 靳岱泽	Responsible for Dashboard, Resources module, Planner module, AI/local suggestion logic, README, and report organization.	50%

10. Conclusion

Smart Learning Assistant solves the problem of scattered information in university students' daily learning management. Through five modules, Dashboard, Schedule, Tasks, Resources, and Planner, the system forms a complete learning management workflow. Users can view courses, manage tasks, save materials, pay attention to deadlines, and generate short-term study plans in the system.

The project reflects user friendliness, immediate feedback, module connection, and interaction optimization. Schedule, Tasks, Resources, Dashboard, and Planner work together, making the system not just a combination of several pages, but an integrated tool designed around students' learning needs.

Although the system still has limitations, such as no backend, no synchronization, no file upload, and no real notification system, it has already implemented a complete demonstrable learning assistant. In the future, if account management, database, notifications, file management, and smarter recommendation logic are added, Smart Learning Assistant can be further expanded into a more practical and personalized learning management platform.